PROTOTYPE SUBMISSION

Saheli

Know before you go.
Get help even when you can't ask for it.

A women's safety companion that shows route risk before a trip, watches the trip with her consent, and escalates on its own if she doesn't check in.

Live: saheli-44mk.onrender.com

GitHub: github.com/neha11igdtuw-ship-it/Saheli

Scan to open the live app

Working prototype · Flask, SQLAlchemy, Twilio, OpenStreetMap · screenshots in this deck are from the running app

PROBLEM STATEMENT

Safety apps wait for an emergency. Most unsafe moments aren't announced.

Women commuting at night, students and late-shift workers in Indian metros make route decisions with no safety information, and rely on an SOS button they may be unable to press.

1 · Reactive by design

SOS apps act after the user triggers them. If she is frightened, restrained or her phone is taken, nothing happens.

2 · No pre-trip insight

Maps rank routes by speed. Nothing tells her that the fastest option crosses a long, dark, isolated stretch.

3 · Fragile delivery

Alerts fail on weak networks and with provider limits. In our own testing, trial SMS accounts silently rejected unverified contacts, exactly when it mattered.

4 · Knowledge stays local

"Don't take that underpass after 9" travels by word of mouth, not to the next person walking the same route.

Problem we solve

How can a woman travelling alone see the risk on her route **before** she leaves, be watched over **during** the trip without continuous tracking, and have help raised **automatically** if she can't act?

HERO FLOW

One journey, end to end

Example: Priya leaves Connaught Place for home in Saket at night.

1

Check the area

Safety Score for her start point, with time-of-day, crime index and community reports itemised.

2

Pick a route

Routes ranked safest-first. A penalty applies to long unlit or low-score stretches.

3

Start a Tracked Trip

ETA 25 min, grace 10 min. Her position is captured once for use if escalated.

4

Check in, or don't

"I've arrived safely" ends it. If she doesn't, a background watchdog notices at ETA + grace.

5

Auto-SOS

Trusted contacts get SMS, call, WhatsApp and email with a map link and nearest help points.

What the user controls

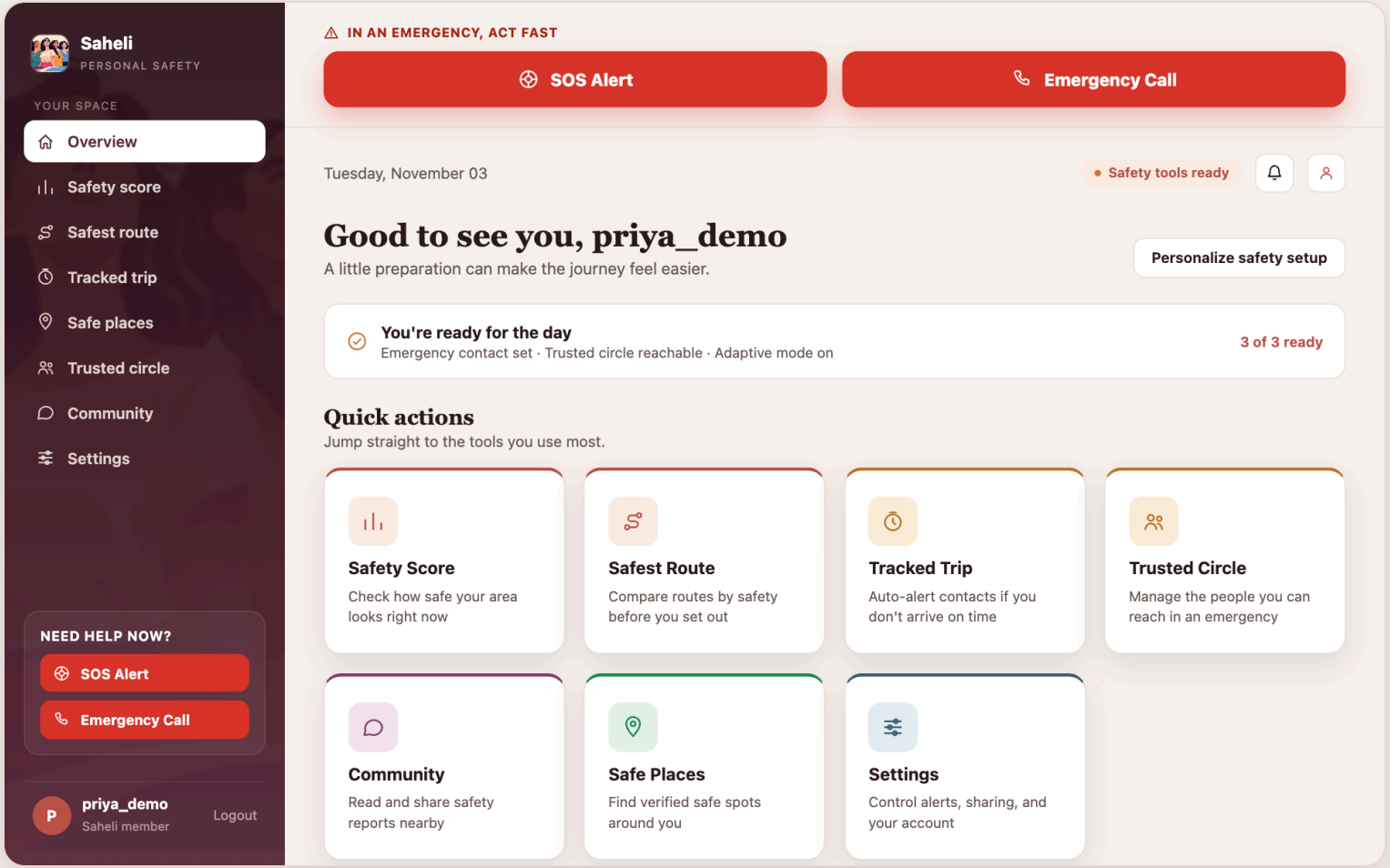
Every escalation passes through the same 10 s cancel window as a manual SOS, so a late check-in doesn't become a false alarm without a chance to stop it.

What happens if the network is down

The SOS page switches to the device's native SMS composer with location (or an honest "no GPS" note) and queues the event for sync when online.

THE PRODUCT

One calm dashboard, emergency actions always visible



SOS and Emergency Call pinned

Visible on every screen, also in the sidebar. Emergency Call dials 112 (India's ERSS).

Readiness check

"3 of 3 ready" confirms emergency contact set, trusted circle reachable and adaptive mode on, so gaps are found before the trip, not during it.

Seven tools, one login

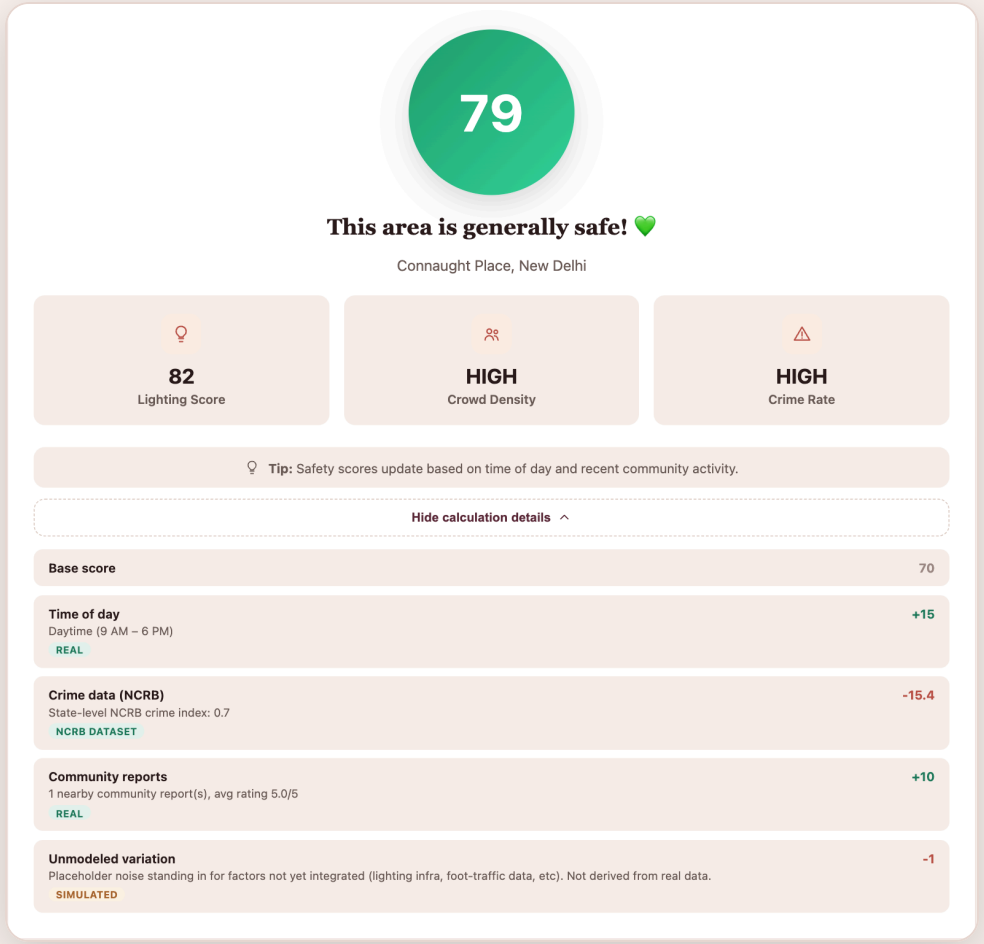
Safety score, safest route, tracked trip, safe places, trusted circle, community, settings.

Screenshot: running prototype, demo account

SAFETY SCORE

A score that shows its working

Every adjustment is itemised and labelled by where it came from. Nothing modelled is presented as measured.



Formula (v0)

`score = 70 + time + crime + community + noise`, clamped to 0 to 100

Labels on screen

REAL time of day, community reports

NCRB DATASET state-level crime index

SIMULATED placeholder noise of ±10 where data isn't integrated yet

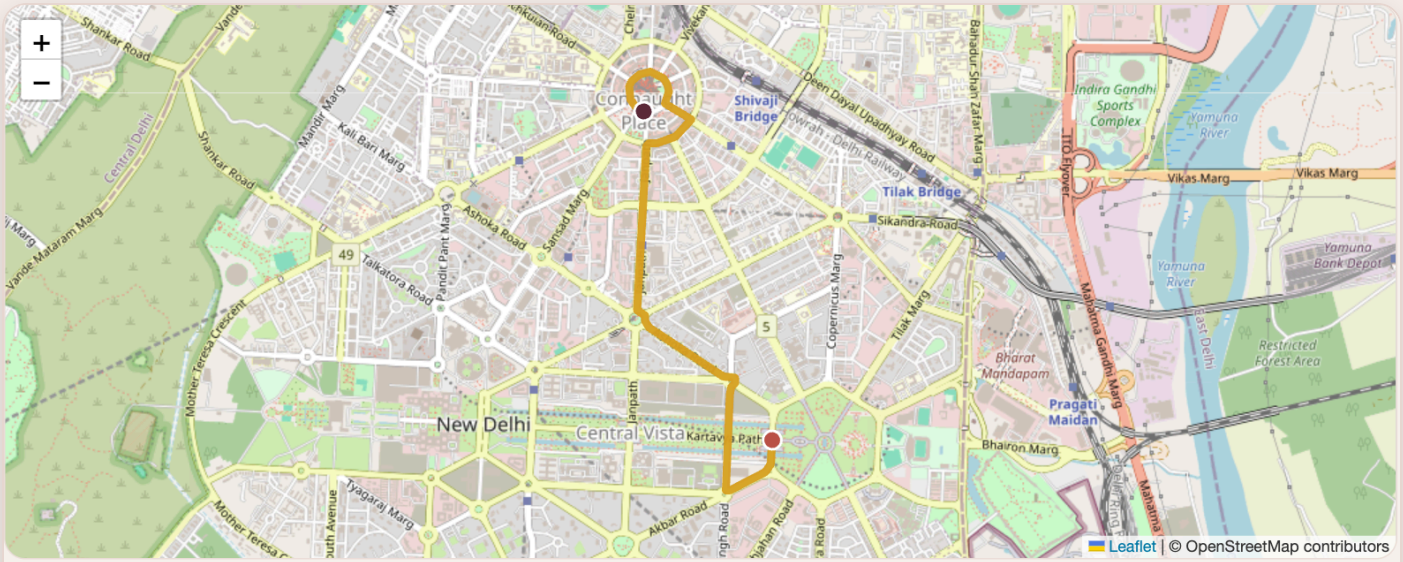
Why this matters

Judges and users can see which part is evidence and which is a placeholder. Next step is removing the placeholder, not hiding it.

Screenshot: Connaught Place, New Delhi, daytime check · demo build

SAFEST ROUTE

Routes ranked by safety, not just speed



Routes, safest first

Scores reflect day conditions. Click a route to view it on the map.

Route 1 · via Route 1 Safest Fastest **74.9/100**

50 mins · 4.2 km

Quickest option.

Save as commute route

How a route is scored

- Route is sampled every **250 m**; each sample maps to a grid cell
- Cell scores are cached **7 days** per time bucket (night, evening, morning, day)
- Route score = length-weighted mean of cells
- **Isolation penalty** up to 25 pts when 3 or more consecutive samples (~750 m) score below 50 or lighting below 40

Engineering detail

- Cold cells scored in parallel (thread pool), since the cost is network IO
- Walking time recomputed at 1.4 m/s: the public OSRM server only returns car-speed timings

Saved commutes

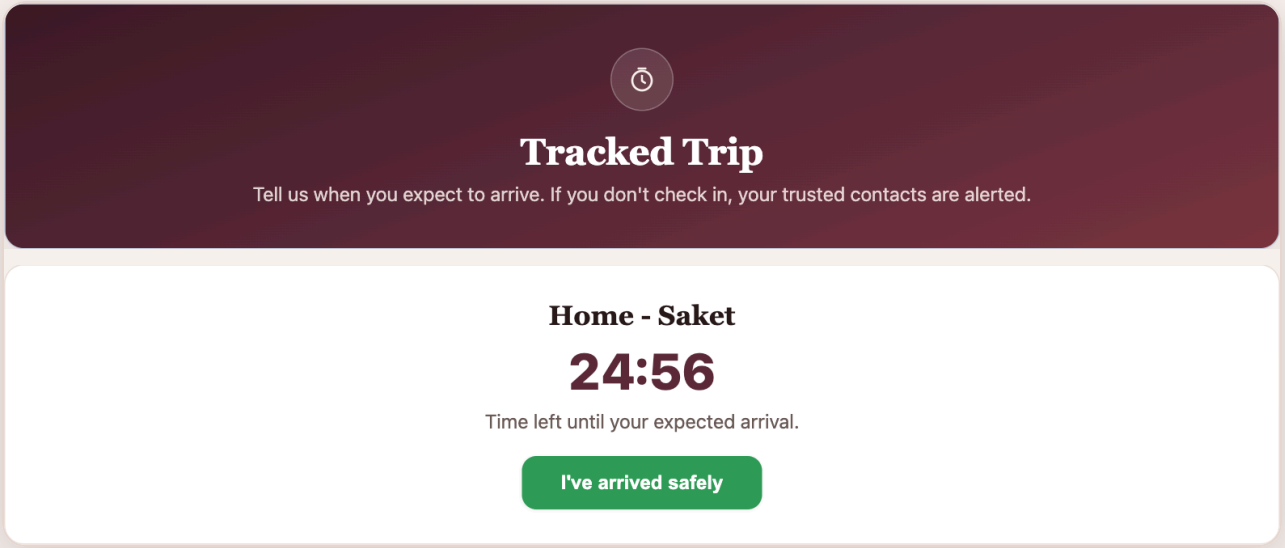
Save 2 routes. A new unsafe or incident report within 300 m triggers an email, at most once per 6 h per route.

Screenshot: Connaught Place to India Gate, walking · OpenStreetMap routing

TRACKED TRIP

Silence becomes the signal

No continuous tracking. The user declares a trip; the server watches the clock.



Mechanics

- ETA from 1 min to 24 h, grace 1 to 60 min (default 10)
- Server watchdog sweeps every 30 s and on every page load or status poll
- **Atomic claim**: a trip escalates exactly once, even if several workers race
- On startup, overdue trips are recovered, so a server restart can't swallow an escalation

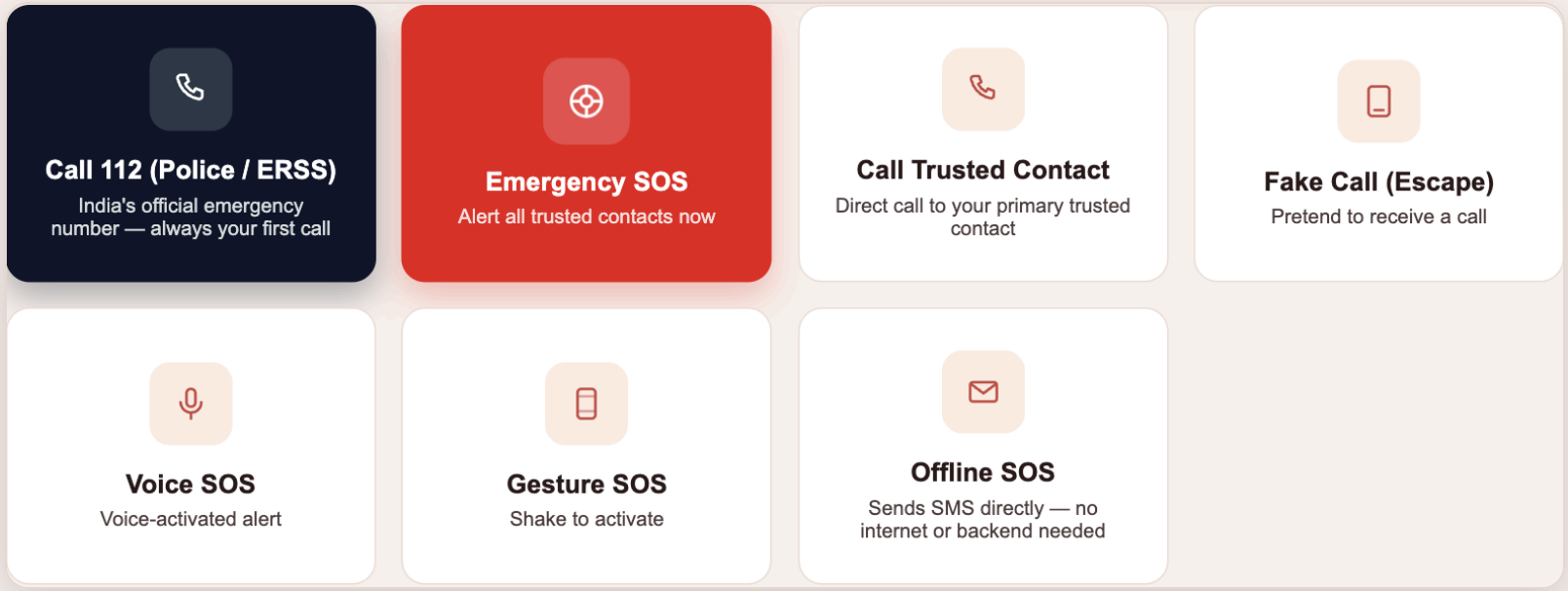
What contacts receive

An alert marked as a missed check-in, last known position, and a map link. If no location was captured, the alert says so instead of implying one.

Screenshot: active trip with live countdown · demo build

EMERGENCY SOS

Built for the failure cases



Cancel window

SOS arms for 10 s. "I'm safe" cancels before anyone is contacted.

Parallel dispatch

Email, SMS, call and WhatsApp fire concurrently per contact, so total latency is the slowest channel, not the sum.

Honest status

Per-channel results are stored and shown. The UI never says "sent" before anything was attempted.

Self-healing

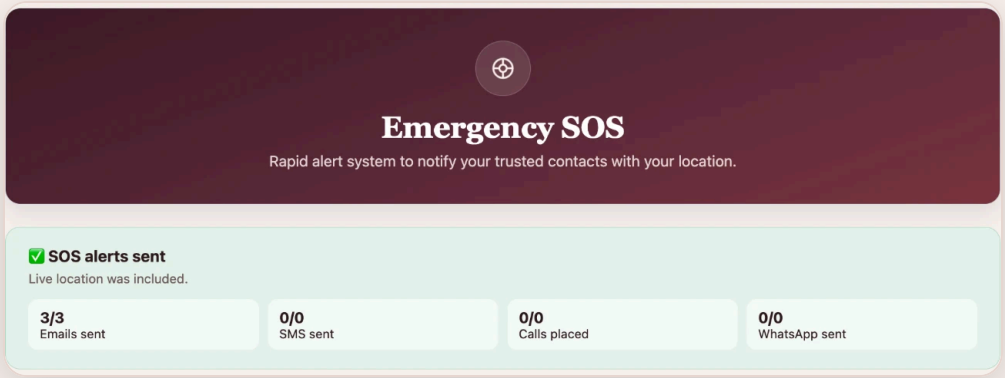
Startup sweep recovers alerts stuck in pending. Unverified numbers on a trial SMS account are skipped up front and surfaced in Trusted Circle.

Triggers: button, phone shake (motion sensor), automatic from Tracked Trip · nearest beacons within 10 km attached to every alert

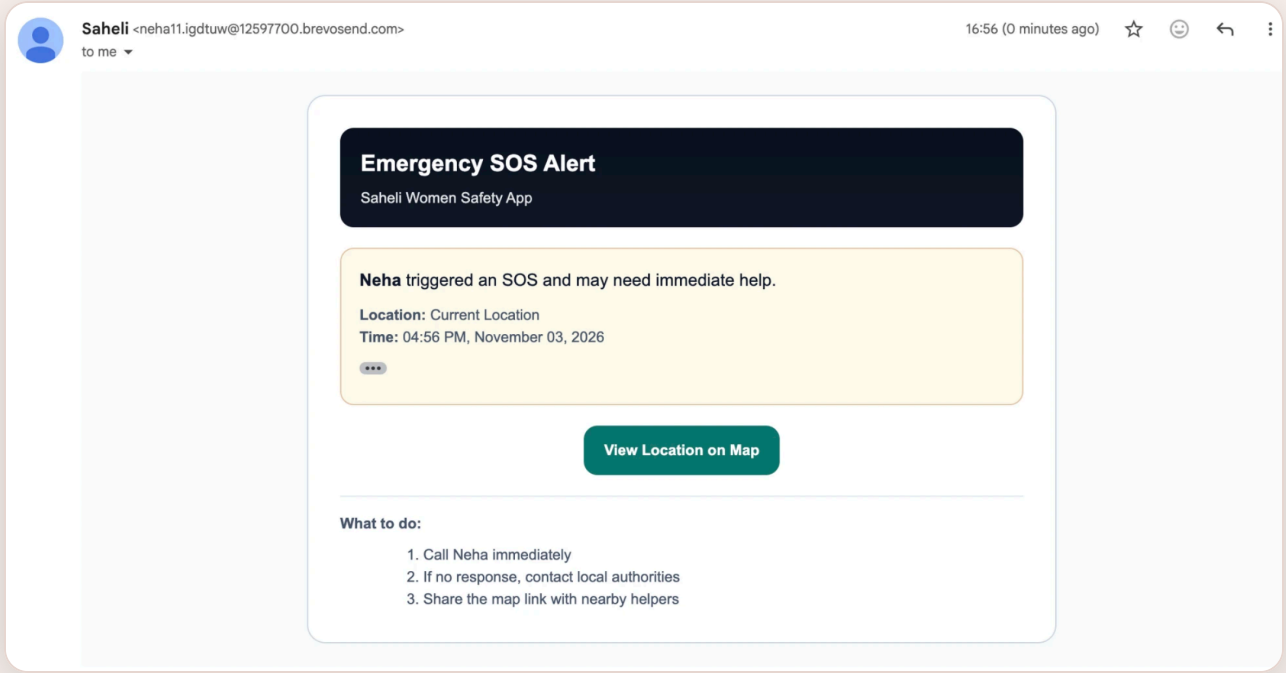
DELIVERY PROOF

A real SOS, delivered to a real inbox

Triggered from the running app with live email settings. Left: the app's own per-channel result. Right: the alert as received at 16:56.



3/3 emails sent. SMS, calls and WhatsApp show 0/0: none were attempted for these contacts (no phone number, or number not verified on a trial SMS account).



Sent via Brevo

Sender shows the Brevo sending domain; delivered to the Inbox, not spam.

What the contact sees

Who triggered it, location, time, a "View Location on Map" button and three next steps.

Honest status

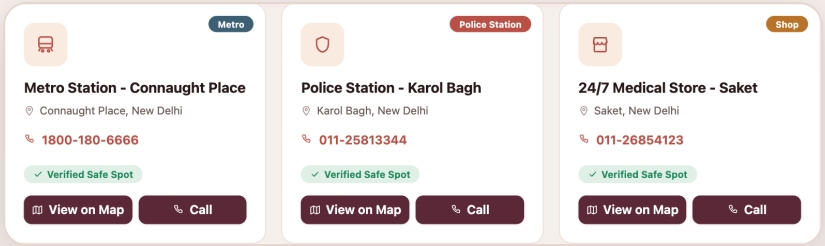
The app reports "sent" only after the provider accepted the message.

Screenshots: Saheli SOS page and Gmail inbox · demo account

COMMUNITY AND SAFE PLACES

Local knowledge feeds the score and warns the next commuter







Reports adjust scores

Safe, unsafe or incident reports with a 1 to 5 rating. Nearby average moves the score by (rating – 3) × 5 points.

Abuse limits

The reporter is never notified of their own report and nothing identifying them is emailed. Per-route 6 h cooldown caps alert bursts.

Safety beacons

Verified metro, police, hospital and 24/7 spots with call and map actions. Four seeded Delhi examples today.

Fake call

Instant, or scheduled for up to 60 min, with a realistic Indian number format.

Screenshots: Community Reports, Safety Beacons, Fake Call · demo build

DATA AND MODEL TRANSPARENCY

What is real, what is a placeholder, what replaces it

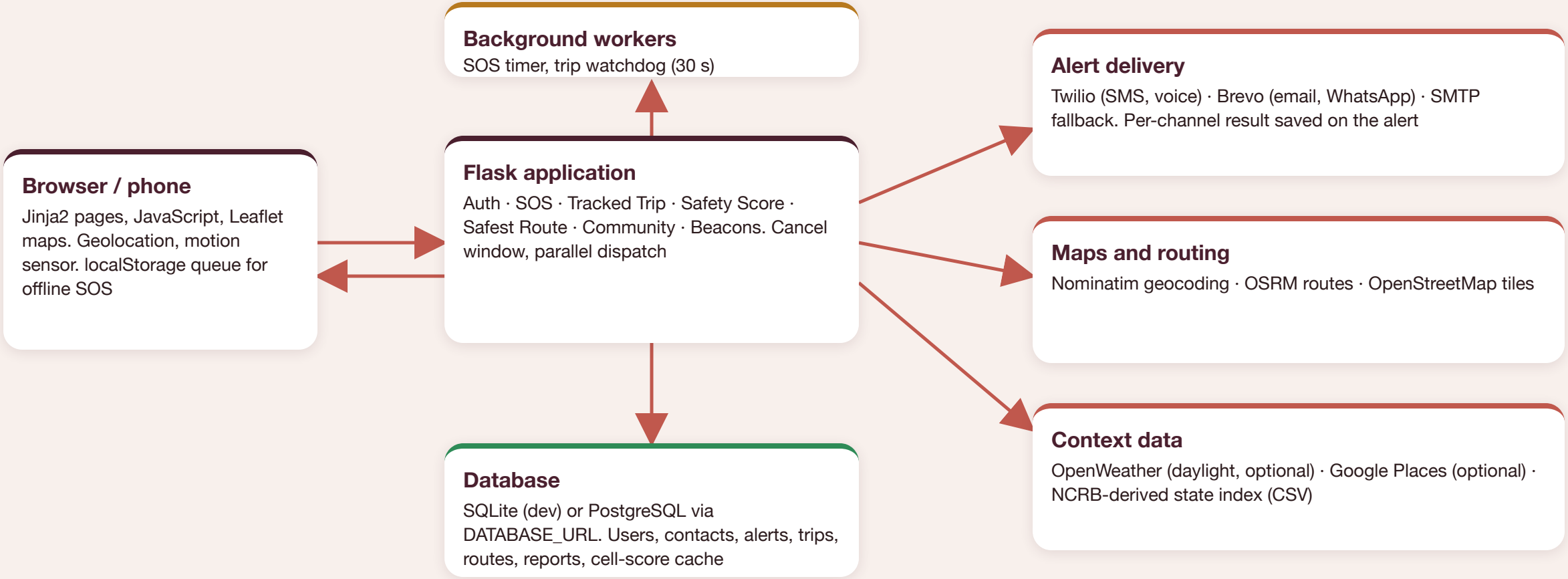
Saheli v0 is a transparent heuristic risk model, not a trained ML model. We say so in the product and here.

Signal	Status today	How it works now	Planned upgrade
Time of day	REAL	Rule-based: night –25, evening –12, morning +8, daytime +15	Calibrate weights against incident timing
Crime context	DATASET	State-level NCRB-derived index (0 to 1) applied as up to –22 pts. Sample file covers 10 states	District-level NCRB data; city police open data
Community reports	REAL	User reports within ~1 km shift score by (avg rating – 3) × 5	Reputation weighting, moderation, decay by age
Lighting	PARTIAL	Daylight and weather via OpenWeather when a key is set; night values estimated	OpenStreetMap street-lamp tags, municipal lighting data
Crowd density	PARTIAL	Google Places open-now ratio when keyed; time-based estimate otherwise	Foot-traffic and POI density from OSM
Variation term	SIMULATED	Random ±10 placeholder, labelled on screen	Remove once the signals above are live

Source: app.py scoring functions and data/ncrb_state_crime_index.sample.csv

ARCHITECTURE

Client, Flask, database, and delivery providers



Deployed with gunicorn. Tables initialise at import time, so a fresh WSGI deploy works without a manual step.

SECURITY Q&A

Safeguards in place, and when we close each gap

Already in the prototype

- Location is sent only when an SOS or escalation fires, or when the user checks a place
- Adaptive mode limits tracking to risky contexts
- Passwords hashed (Werkzeug); email and password validation
- Input bounds on ETA, grace and fake-call delay; rejected rather than silently defaulted
- User text HTML-escaped in outgoing emails
- Reporter anonymity and email-burst cooldowns
- Secrets read from environment variables, not code

Known gaps and when we close them

- **CSRF protection and rate limiting** (SOS, login, reports): **first 30 days, before any pilot with real users**
- **Report moderation** and reputation: first 30 days
- **Contact opt-in** and retention limits (DPDP Act, 2023): first 30 days
- **Durable job queue** for SOS timers: 30 to 90 days, with PostgreSQL deployment
- **Encryption at rest** for contacts and location history: 30 to 90 days
- False-alarm tracking: during the pilot

TEAM

Who built what

Neha

2nd year, B.Tech CSE · IGDTUW

UI and frontend | Screens and theme | Offline SOS interface

Designed and built the screens: dashboard, safety score, route, trip, SOS, community and beacons.

Varsha

SE-1 · Zinnia Pvt. Ltd.

Backend | Integrations | Reliability

Built the Flask APIs, scoring and route engine, trip watchdog, parallel SOS dispatch, and the Twilio and Brevo integrations.

ROADMAP

From transparent heuristic to trusted safety network

Next 30 days

- Remove the simulated noise term
- District-level NCRB index and OSM street-lamp data
- Report moderation, rate limits, CSRF
- Contact opt-in confirmation

30 to 90 days

- Campus pilot, with time-to-notify, delivery rate and false-escalation targets
- Durable job queue and PostgreSQL deployment
- Hindi and regional-language UI (language field already in user profile)
- Installable mobile web app (PWA)

90 days and beyond

- Calibrate the score on pilot outcomes; move from heuristic to a validated model
- Voice-distress and wearable triggers
- Partner dashboards for campuses and employers

Saheli: safety that works even when she can't ask. Demo path: check score → compare routes → start a trip → let it lapse → watch the auto-SOS arm with its cancel window.

Try it (scan the QR): saheli-44mk.onrender.com · **Code:** github.com/neha11igdtuw-ship-it/Saheli